Exam Problem 1 • High Difficulty

CIRCONSA – Semester Financial Statements & Cash Reconciliation

Original statement · T3_Additional_EXERCISES.pdf

The financial director of **CIRCONSA** is working on its financial statements (Balance Sheet and Income Statement) as of **30 June 2015 (30/06/2015)**. Starting from the Balance Sheet at 31/12/2014 (Current Assets: 5,400 [Cash 1,200, Receivables 1,800, Stock 2,400]; Non-Current Assets: Machinery 500 less Accum Dep 50 = 450; Current Liabilities: 800 [Taxes 350, Suppliers 450]; Owners' Equity: 5,050 [Retained Earnings 850, Capital 4,200]; Total = 5,850 m.u.), evaluate the semester operations from 01/01/2015 to 30/06/2015:

- **i. Sales:** 1,200 units at 20 mu/u (200 u/month). 30-day payment term. VIP client (30% of sales) receives a 10% commercial discount.
- **ii. Purchases:** 1,000 units (40% in Q1, 60% in Q2) at 12 mu/u. 90-day supplier payment term.
- **iii. Initial Stock:** 200 units at 12 mu/u (2,400 mu).
- **iv. Prior Year Liabilities:** Q1 2015 paid 2014 unpaid taxes (350 mu) and supplier balance (450 mu).
- **v. Prior Year Receivables:** Q1 2015 collected 2014 customer debt except 300 mu deferred to July 2016.
- **vi. Operating Expenses:** Personnel 350 mu/mo + Social Security 150 mu/mo (paid in month produced); Overheads 300 mu/mo (paid 2 months after).
- **vii. Renting:** 1 April 2015 computer renting for 400 mu/month, payment term 120 days.
- **viii. Patent:** 1 April 2015 purchased patent for 2,400 mu, payable 1 July 2015. 4-year linear amortization.
- **ix. Bank Credit:** 1 Feb 2015 borrowed 2,000 mu at 12% per annum, principal + interest due 1 Feb 2016.
- **x. Machine 1:** Bought 1 July 2014, amortized linearly with 20% residual value.
- **xi. Capital & Machine 2:** 1 Feb 2015 cash capital increase 10,000 mu. Same date bought Machine 2 for 12,000 mu (paid 8,000 mu cash, 4,000 mu payable 1 July 2015; 4-year life, 0 residual value).
- **xii. Corporate Tax:** 20% tax, recorded as "taxes to pay" on 30 June 2015.

EXAM DAY STRATEGY: TIME HORIZONS & DEFERRED PAYABLES

- **Semester Duration:** $t = 6$ months $= 0.5$ year. Multiply monthly figures by 6 (or active months).
- **Payable on 1 July 2015:** The patent debt (2,400 mu) and Machine 2 debt (4,000 mu) are payable *the day after* the balance date. Therefore, they are **Current Liabilities on 30/06/2015**, not cash outflows!
- **Customer Deferral:** The 300 mu deferred to July 2016 is > 1 year away $\implies$ **Non-Current Account Receivable**.
- **Payment Terms:** 30-day customer delay means June sales are unpaid on 30 June; 90-day supplier delay means all Q2 purchases are unpaid on 30 June.

Analytical Deductions Step-by-Step

1. **Net Sales:** Total gross sales $= 1,200 \times 20 = 24,000$ mu. VIP client ($30\% = 7,200$ mu) discount of $10\% = 720$ mu.

$$\text{Net Sales} = 24,000 - 720 = \mathbf{23,280} \text{ mu}$$

   June sales ($1/6$ of total) $= 3,880$ mu are uncollected on 30/06 $\implies$ **Current Accounts Receivable** $=$ $\mathbf{3,880}$ mu. Collections $= 23,280 - 3,880 = \mathbf{19,400}$ mu.

2. **COGS & Ending Inventory:** Available units $= 200$ init $+ 1,000$ purch $= 1,200$ units. Sold $=$ $1,200$ units $\implies$ **Ending Inventory** $= \mathbf{0}$ mu.

$$\text{COGS} = 1,200 \times 12 = \mathbf{14,400} \text{ mu}$$

   Q2 purchases (600 units $\times 12 = 7,200$ mu) payable at 90 days are unpaid on 30 June $\implies$ **Accounts Payable (Suppliers)** $= \mathbf{7,200}$ mu.

3. **Personnel, Overheads & Renting:**
   - Personnel: 6 months $\times (350 + 150) = \mathbf{3,000}$ mu (paid in full).
   - Overheads: 6 months $\times 300 = \mathbf{1,800}$ mu. May and June (600 mu) unpaid $\implies$ **Overheads Payable** $= \mathbf{600}$ mu.
   - Renting (Apr–Jun, 3 months): $3 \times 400 = \mathbf{1,200}$ mu. Payable at 120 days $\implies$ **Renting Debt** $=$ $\mathbf{1,200}$ mu.

4. **Depreciation & Amortization:**
   - Patent ($2,400/48$ mo $\times 3$ mo): $\mathbf{150}$ mu (Net BV $= 2,250$ mu).
   - Machine 1 ($500 - 100$ res val $= 400$ base over 4 yrs, semester dep): $\mathbf{50}$ mu (Accum dep $= 100$ mu). The 4-year life follows from the opening accumulated depreciation: 50 mu over the 6 months since 1 July 2014 gives 100 mu/year, so life $= 400/100 = 4$ years.
   - Machine 2 ($12,000/48$ mo $\times 5$ mo active): $\mathbf{1,250}$ mu (Accum dep $= 1,250$ mu).
   - Total Depreciation in P&L: $150 + 50 + 1,250 = \mathbf{1,450}$ mu.

5. **Bank Credit Interest:** $2,000 \times 12\% \times \frac{5}{12} = \mathbf{100}$ mu (Accrued interest payable $= 100$ mu).

6. **EBIT, Tax & Net Income:**

$$\text{EBIT} = 23,280 - 14,400 - (3,000 + 1,800 + 1,200 + 1,450) = 8,880 - 7,450 = \mathbf{1,430}\ \text{mu}$$

$$\text{EBT} = 1,430 - 100 = \mathbf{1,330}\ \text{mu}$$

$$\text{Corporate Tax } (20\%) = 20\% \times 1,330 = \mathbf{266}\ \text{mu} \implies \text{Net Profit} = 1,330 - 266 = \mathbf{1,064}\ \text{mu}$$

Balanced Financial Statements at 30 June 2015

ASSETS (Activo)	Amount (mu)	LIABILITIES & EQUITY (Pasivo y PN)	Amount (mu)
NON-CURRENT ASSETS	**13,700**	**STOCKHOLDERS' EQUITY**	**16,114**
• Patent (Net Book Value: 2,400 - 150)	2,250	• Paid-in Capital (4,200 + 10,000)	14,200
• Machinery (Net BV: 12,500 - 1,350)	11,150	• Retained Earnings (2014)	850
• Long-term Customer Debt (July 2016)	300	• Net Profit for Semester	1,064
CURRENT ASSETS	**20,180**	**CURRENT LIABILITIES**	**17,766**
• Inventories (Stock)	0	• Short-term bank credit	2,000
• Accounts Receivable (June sales)	3,880	• Accrued loan interest	100
• Cash & Bank Deposits (Reconciled)	16,300	• Accounts payable (Suppliers Q2)	7,200
		• Debt for Machine 2 (due 1/7)	4,000
		• Debt for Patent (due 1/7)	2,400
		• Overheads payable (May & June)	600
		• Renting debt payable (Apr–Jun)	1,200
		• Taxes to pay (20% Corporate Tax)	266
TOTAL ASSETS	**33,880**	**TOTAL LIABILITIES & EQUITY**	**33,880**

Cash derivation (30/06/2015): inflows $1,200 + 1,500 + 19,400 + 2,000 + 10,000 = 34,100$ mu; outflows $350 + 450 + 4,800 + 3,000 + 1,200 + 8,000 = 17,800$ mu (2014 taxes, 2014 suppliers, Q1 purchases 400×12, personnel, overheads of January–April, Machine 2 down payment); cash $= 34,100 - 17,800 = \mathbf{16,300}$ mu.

EQUILIBRIUM VERIFICATION

$$\text{Assets } (\mathbf{33,880}\ \text{mu}) \equiv \text{Liabilities } (\mathbf{17,766}\ \text{mu}) + \text{Equity } (\mathbf{16,114}\ \text{mu}) = \mathbf{33,880}\ \text{mu}$$

Four-Phase Audit Trail (journal entries and ledger)

Phase 1: Hypotheses. Period 1 Jan to 30 Jun 2015 (6 months), accrual basis. Sales are uniform (200 u/month); the 30-day term leaves only June sales open; purchases are paid 90 days later, so Q1 purchases are settled by 30 June and all Q2 purchases remain open. Machine 1's 4-year life is inferred from the opening accumulated depreciation (50 over 6 months gives 100/year on a base of 500 - 100 = 400). Interest is simple (months/12); tax is 20% of EBT, left as "taxes to pay".

Phase 2: Definitions. $A = L + E$ with $E = \text{Paid-in capital} + \text{Retained earnings}_{2014} + \text{Net income}$. Assets carry debit balances; liabilities and equity carry credit balances; income-statement accounts are closed into equity.

Phase 3: Journal entries (m.u.). Each entry was generated and balanced in Python (debits = credits).

#	Event	Debit	Credit
1	Opening balance 31/12/2014 (A = L + E: 5,850 = 800 + 5,050)	Cash 1,200 Accounts receivable 1,800 Inventory 2,400 Machinery 500	Accum. depreciation 50 Taxes payable 350 Suppliers payable 450 Retained earnings 850 Paid-in capital 4,200
2	(i) Sales 1,200 u x 20 less 10% on the 30% VIP share	Accounts receivable 23,280 Sales discount 720	Sales 24,000
3	(i) Collection of Jan-May sales (30-day term; June stays open)	Cash 19,400	Accounts receivable 19,400
4	(ii) Purchases 1,000 u x 12	Inventory 12,000	Suppliers payable 12,000
5	(ii) Payment of Q1 purchases (400 u x 12, 90 days)	Suppliers payable 4,800	Cash 4,800
6	(iii) COGS = 2,400 opening + 12,000 purchases - 0 ending stock	COGS 14,400	Inventory 14,400
7	(iv) Payment of 2014 taxes and suppliers	Taxes payable 350 Suppliers payable 450	Cash 800
8	(v) Collection of 2014 receivables, 300 deferred to July 2016	Cash 1,500 Long-term receivable 300	Accounts receivable 1,800
9	(vi) Personnel + Social Security 6 x (350 + 150), paid in the month	Personnel 3,000	Cash 3,000
10	(vi) General expenses 6 x 300; Jan-Apr paid, May-Jun open	Overheads 1,800	Cash 1,200 Overheads payable 600
11	(vii) Computer renting 3 x 400 (term 120 days, nothing due yet)	Renting 1,200	Renting payable 1,200
12	(viii) Patent purchased 1 April, payable 1 July	Patent 2,400	Patent payable 2,400
13	(viii) Patent amortization 2,400/48 x 3	Amortization 150	Accum. amortization 150
14	(ix) Bank credit received 1 February	Cash 2,000	Bank credit 2,000
15	(ix) Accrued interest 2,000 x 12% x 5/12	Interest expense 100	Accrued interest 100
16	(x) Depreciation machine 1: (500 - 100)/4 x 6/12	Depreciation 50	Accum. depreciation 50
17	(xi) Capital increase 1 February	Cash 10,000	Paid-in capital 10,000
18	(xi) Machine 2: 12,000 with 8,000 cash, 4,000 payable 1 July	Machinery 12,000	Cash 8,000 Machine payable 4,000
19	(xi) Depreciation machine 2: 12,000/48 x 5	Depreciation 1,250	Accum. depreciation 1,250
20	(xii) Corporate tax 20% x 1,330	Tax expense 266	Taxes payable 266

Closing ledger account	Balance (m.u.)
Cash	16,300 Dr
Accounts receivable	3,880 Dr
Long-term receivable	300 Dr
Inventory	0 Cr
Patent	2,400 Dr
Accum. amortization	150 Cr
Machinery	12,500 Dr
Accum. depreciation	1,350 Cr
Suppliers payable	7,200 Cr
Overheads payable	600 Cr
Renting payable	1,200 Cr
Patent payable	2,400 Cr
Machine payable	4,000 Cr
Bank credit	2,000 Cr
Accrued interest	100 Cr
Taxes payable	266 Cr
Paid-in capital	14,200 Cr
Retained earnings	850 Cr

Phase 4: Units and sanity checks. Currency: m.u. Accounting equation: $33{,}880 = 17{,}766 + 16{,}114$. Cash identity: $1{,}200 + 34{,}100 - 17{,}800 = 16{,}300$, equal to the ledger cash balance. Limit check: if no June sale were open, receivables would be 0 and cash 3,880 higher while profit would stay the same (accrual): receivables move cash, not profit. No official numerical key is available in `sources/` (only the statement PDF), so the figures stand on the recomputation above.

`Study notes written from the official course material (T3_Additional_EXERCISES.pdf; Topic 3 Financial Management (I).pdf). Online version: subjects/business-management/problemas/topic-3-extra-exam-problems.html`